

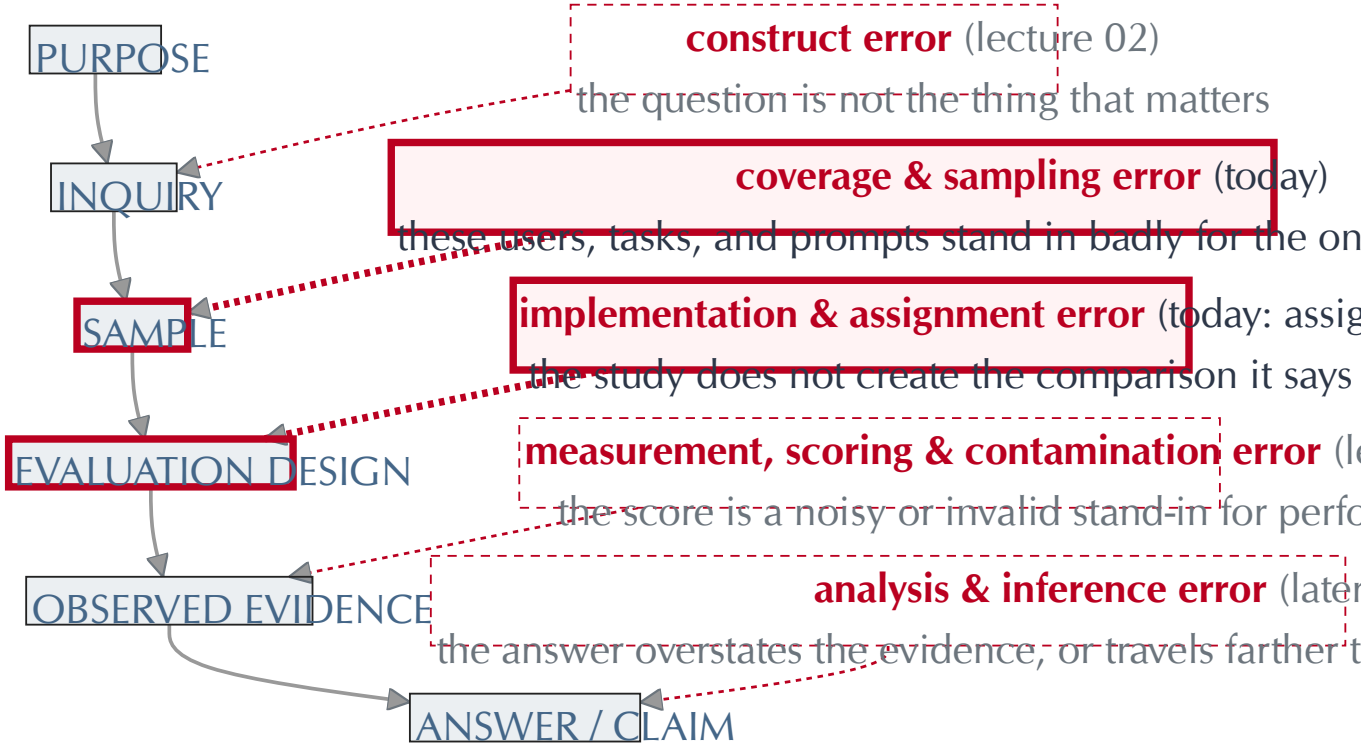
# Experimental Design for AI Evaluation

Session 5

2026-09-29

# This week's links in the total error chart

Last week: valid scores. Today: **which units** and **which comparisons**.



# Simulation and power

# Test retrieval's effect on learning

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- **Outcome:** points on a later **unaided quiz of new problems**, using last week's rubric.

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- **Hold fixed:** practice problems, model, generic instruction, and quiz.

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Can this study reliably detect a **3-point improvement in unaided quiz scores**?

# Simulate the main features of your outcome

- **Outcome:** the unaided quiz score, on a 0–100 scale.
- **Student variation:** students differ in expected quiz performance.
- **Measurement noise:** question selection and scoring add variation.
- **Treatment effect:** retrieval during practice may shift expected performance.

# Simulate the main features of your outcome

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- The tuning demo models individual ratings of model answers at a specified annotator reliability.

# Simulate learning and measurement separately

$$\text{Quiz score} = \text{expected performance} + \text{treatment effect} + \text{noise}.$$

- Retrieval shifts **expected unaided performance** by 3 points.

```
N <- 200
model <- declare_model(
  N = N, T0 = rnorm(N, mean = 50, sd = 8),
  error = rnorm(N, sd = 6),
  potential_outcomes(Y ~ T0 + 3 * Z + error)
)
inquiry <- declare_inquiry(
  ATE = mean(Y_Z_1 - Y_Z_0)
)
```

More measurement noise makes the same effect harder to detect.

# Declare how the study produces an answer

```
assignment <- declare_assignment(
  Z = complete_ra(N, m = N / 2)
)
measurement <- declare_measurement(
  Y = reveal_outcomes(Y ~ Z)
)
estimator <- declare_estimator(
  Y ~ Z, .method = lm_robust, inquiry = "ATE"
)
design <- model + inquiry + assignment +
  measurement + estimator
```

Complete randomization fixes 100 people per Bundle. `lm_robust` estimates their mean difference with a robust standard error.



# One simulated study produces one answer

```
set.seed(2)
one_study <- draw_data(design)
one_answer <- estimator(one_study)
```

| Estimate | SE | Lower 95% | Upper 95% |
|---|---|---|---|
| 5.402 | 1.464 | 2.516 | 8.288 |

- The true effect in our model is 3 points.
  - Random assignment and the sample determine the answer from this particular study.



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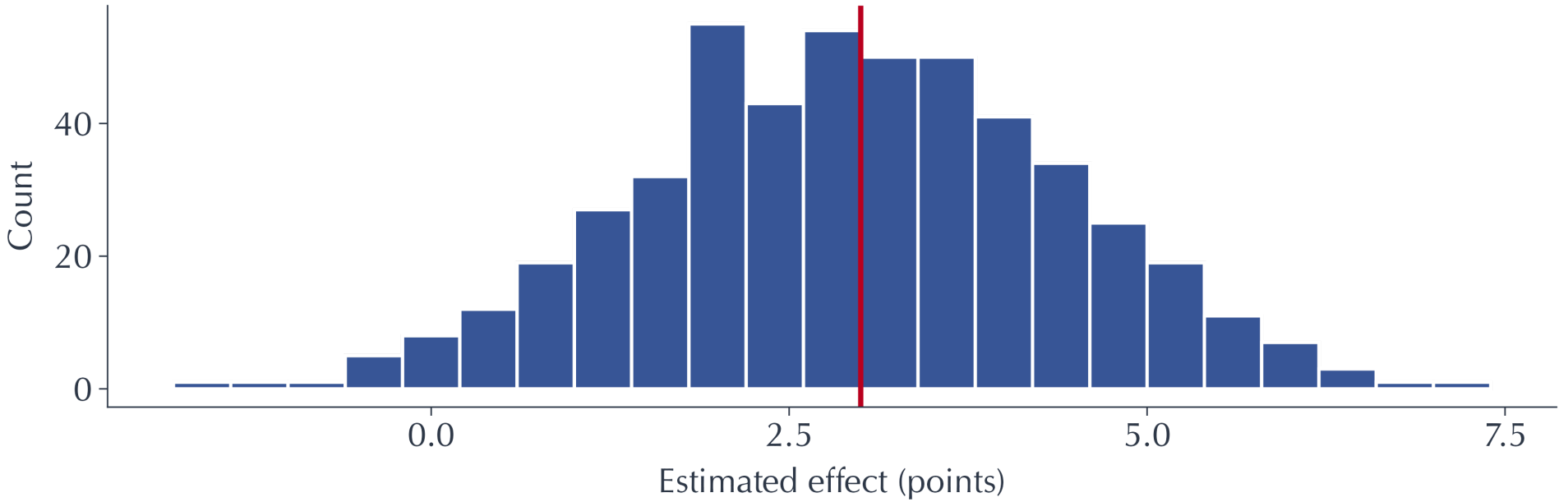
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- The true effect in our model is 3 points.
  - Random assignment and the sample determine the answer from this particular study.
- Repeating the full study shows how much that answer varies.



# Repetition reveals uncertainty

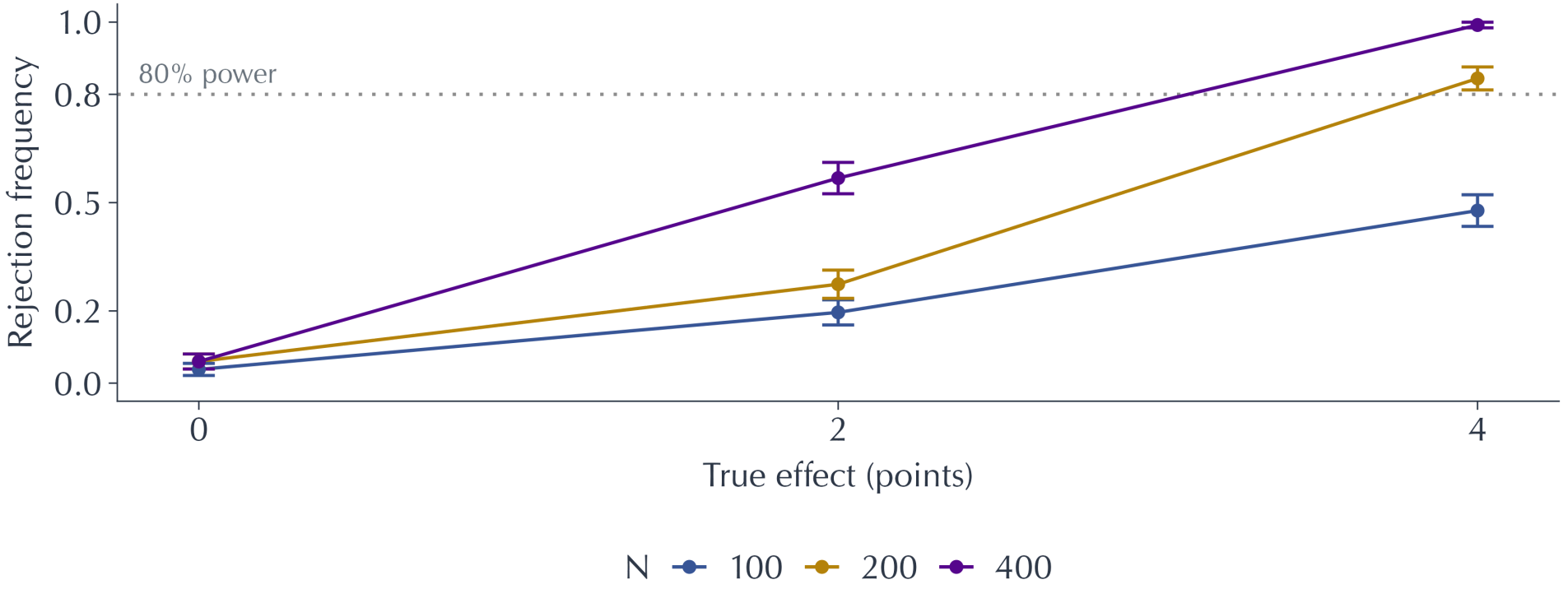
```
diagnosis <- diagnose_design(design, sims = 500)
get_diagnosands(diagnosis)
```




The red line is the true effect, 3 points. **Bias** is displacement of the average estimate; **precision** is the spread across studies.

# Larger samples help detect smaller effects

**Power** is the fraction rejecting a specified false null. Here: two-sided tests at 5%.




- At effect zero, rejection rate is Type I error.
- Smallest effect size with power $\geq$ 80% is the "minimal detectable effect" (MDE)

# Summary: `DeclareDesign` for power analysis

- **Represent your data:**
  - score variation, differences across people or items, and repeated measurements.
- **Simulate your design and analysis:**
  - sample sizes, treatment assignment, comparisons, and tests.
- **Vary plausible effects and assumptions:**
  - identify which effects you can reliably detect.



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**Your pre-analysis plan (PAP) must discuss power.**

- Given your design and measures, what effect sizes do you expect to detect?
- Report power or an MDE for your main contrast, in meaningful units, with the assumptions behind it.



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Worked code and tuning steps: DeclareDesign walkthrough · Runnable QMD.



# Factorial experiments

# Vary components of the Bundle

Every condition, including the control, specifies settings in the Bundle.

| Bundle component | Possible variation |
| --- | --- |
| Model weights | Smaller vs. larger model |
| Decoding | Lower vs. higher temperature |
| **Instruction (prompt)** | Generic assistance vs. hint-first tutoring |
| **Retrieval (harness)** | Course-note access off vs. on |
| Interface | Chat window vs. hints beside the problem |
| Workflow | Help available immediately vs. after an attempt |

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We will cross **instruction** and **retrieval**, holding the other components fixed, to separate their effects and interaction.



# Cross two components to create four Bundles

During practice, vary course-note retrieval $R$ and hint-first instruction $H$.

| | Generic instruction: $H = 0$ | Hint-first instruction: $H = 1$ |
|---|---|---|
| Retrieval off: $R = 0$ | $B_{00}$ | $B_{01}$ |
| Retrieval on: $R = 1$ | $B_{10}$ | $B_{11}$ |

- **Hold fixed:** the unaided quiz and all other Bundle components.
- **Reference** $B_{00}$**:** a functioning chatbot with generic instruction and retrieval off.

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- **Hold fixed:** the unaided quiz and all other Bundle components.
- **Reference $B_{00}$:** a functioning chatbot with generic instruction and retrieval off.
- **Expand the comparison:**
  - Before: **retrieval** alone, $B_{10}$ vs. $B_{00}$.
  - Now: cross **retrieval** with **instruction**.
  - Estimate each component's effect and their interaction.

# A linear model makes the contrasts explicit

$$Y_i(r, h) = \mu + a_i + \beta_R r + \beta_H h + \beta_{RH} rh + \epsilon_i.$$

$a_i$: differences in true quiz performance. $\epsilon_i$: quiz and scoring noise.
Set $\mu = 50$, $\beta_R = 3$, $\beta_H = 1$, $\beta_{RH} = 4$.

| Expected quiz score | Generic | Hint-first |
|---|---|---|
| Retrieval off | 50 | 51 |
| Retrieval on | 53 | 58 |

Scores and effects are in points. Baseline SD: 10 points; Gaussian approximation.

# A component effect depends on the other components

| Contrast | Formula | Points |
|---|---|---|
| Retrieval with generic instruction | $\beta_R$ | 3 |
| Retrieval with hint-first instruction | $\beta_R + \beta_{RH}$ | 7 |
| Both on vs. both off | $\beta_R + \beta_H + \beta_{RH}$ | 8 |
| Interaction: change in retrieval effect | $\beta_{RH}$ | 4 |

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- The interaction says that retrieval helps more under hint-first instruction here.
- Larger effects mean more power (all else equal)!

# The AMCE averages over specified Bundle settings

- **Average marginal component effect (AMCE):** average a component's effect over the other components.
  - p: probability of **hint-first instruction** in the target distribution.

$$AMCE_R = \beta_R + p\beta_{RH}.$$

- **Choose the weights:** $p = 1/2$ gives **5 points**; $p = 1/4$ gives **4 points**.

Hainmueller, Hopkins & Yamamoto (2014), Causal Inference in Conjoint Analysis.

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- **Why average?** Pool observations across settings of the other components.
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- **Why average?** Pool observations across settings of the other components.
  - Can improve power relative to comparing just two cells.
  - Especially useful with many components: no need to estimate every cell mean.
- **With independent randomization:** many full cells can be empty in the sample.
  - Both component levels must still be possible at the settings we average over.

Hainmueller, Hopkins & Yamamoto (2014), Causal Inference in Conjoint Analysis.

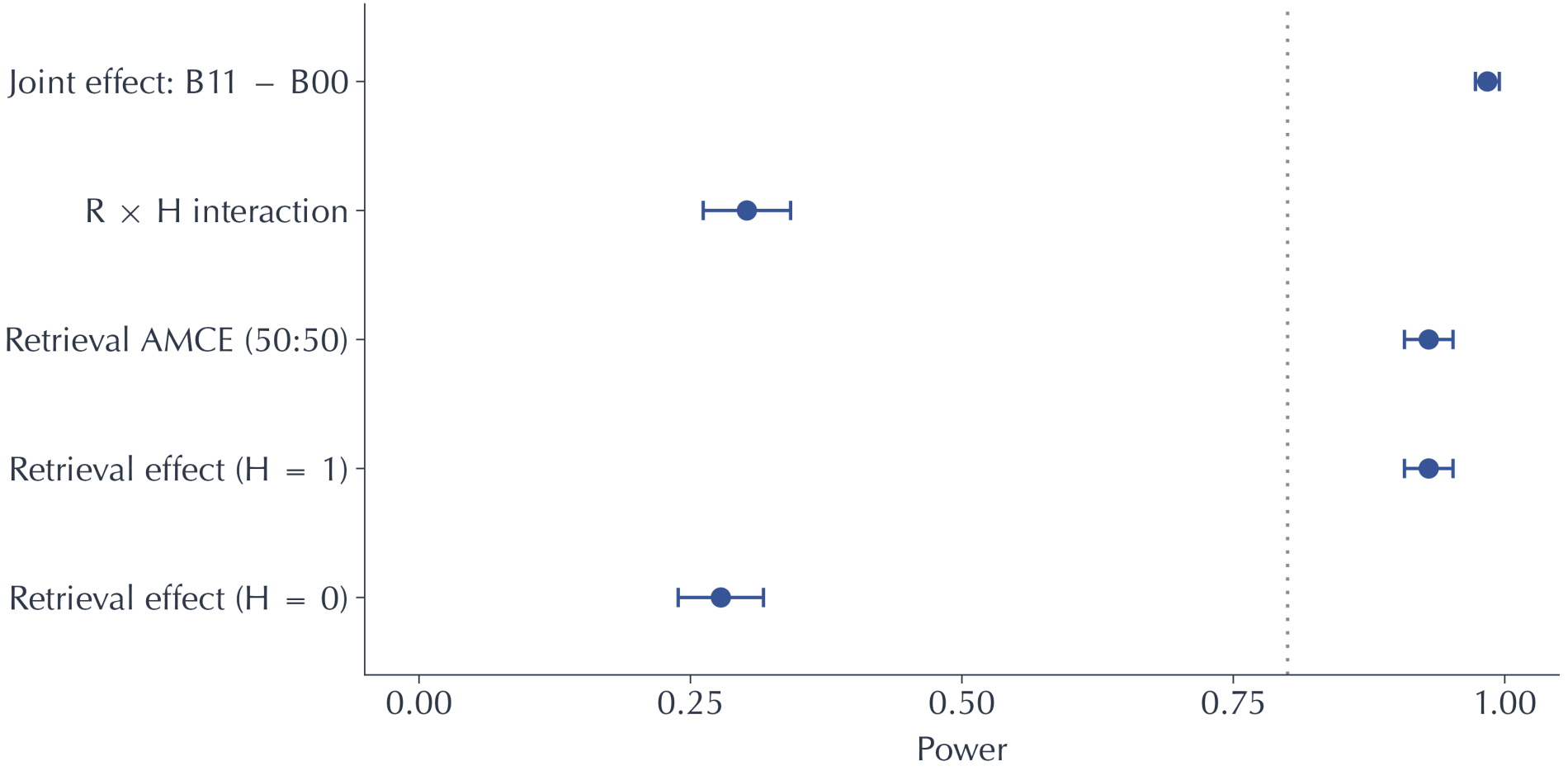


# Expand the declaration to four Bundles

Use the same model–inquiry–data–answer structure.

```
assignment <- declare_assignment(
  cell = complete_ra(N,
    conditions = c("B00", "B10", "B01", "B11"),
    m_each = rep(N / 4, 4))
)
inquiry <- declare_inquiry(
  AMCE_R = mean((Y_B10 - Y_B00 +
                 Y_B11 - Y_B01) / 2),
  interaction = mean(Y_B11 - Y_B01 - Y_B10 + Y_B00)
)
```

# Power differs across contrasts in the same experiment




N = 50 per Bundle. Power depends on each contrast's effect size **and** uncertainty.

# Within versus between designs

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  - Score comparable tasks under each; compare within-person differences.

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  - Randomize order: **A** → **B** or **B** → **A**.
  - Score comparable tasks under each; compare within-person differences.
- **Model evaluations:** pair A and B on the same item, in separate fresh runs.

# Within designs remove shared variation

- **Compare two Bundles:** $c \in \{0, 1\}$.

$$Y_{ic} = \mu + \tau c + a_i + \epsilon_{ic}.$$

- **Difference within a person or item:**

$$D_i = Y_{i1} - Y_{i0} = \tau + \epsilon_{i1} - \epsilon_{i0}.$$

- Shared ability or difficulty ($a_i$) cancels.
- **Assumption:** no carryover between Bundle exposures.

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- **People:** use comparable tasks.
  - Randomize order and task-to-Bundle allocation.
- **Models:** use the same item and scoring rule.
  - Reset context and isolate tool state.

# Positive covariance makes pairing useful

- **In the model:** $a_i$ and the $\epsilon_{ic}$ terms are independent.
  - Shared variation: $Var(a_i) = \sigma_a^2$.
  - Residual variation: $Var(\epsilon_{ic}) = \sigma_\epsilon^2$.

$$V_{within} = \frac{2\sigma_\epsilon^2}{n}, \qquad V_{between} = \frac{2(\sigma_a^2 + \sigma_\epsilon^2)}{n}.$$

- **Same score count:** $2n$.
  - Within: $n$ pairs. Between: $n$ separate units per arm.

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- **Correlation is the shared fraction:** $\rho = \sigma_a^2/(\sigma_a^2 + \sigma_\epsilon^2)$.
  - 80% shared variation: pairing leaves **one fifth** of the variance.
  - No shared variation ($\sigma_a^2 = 0$): no gain.

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  - No shared variation ($\sigma_a^2 = 0$): no gain.
- Recruitment costs and participant burden may still differ.

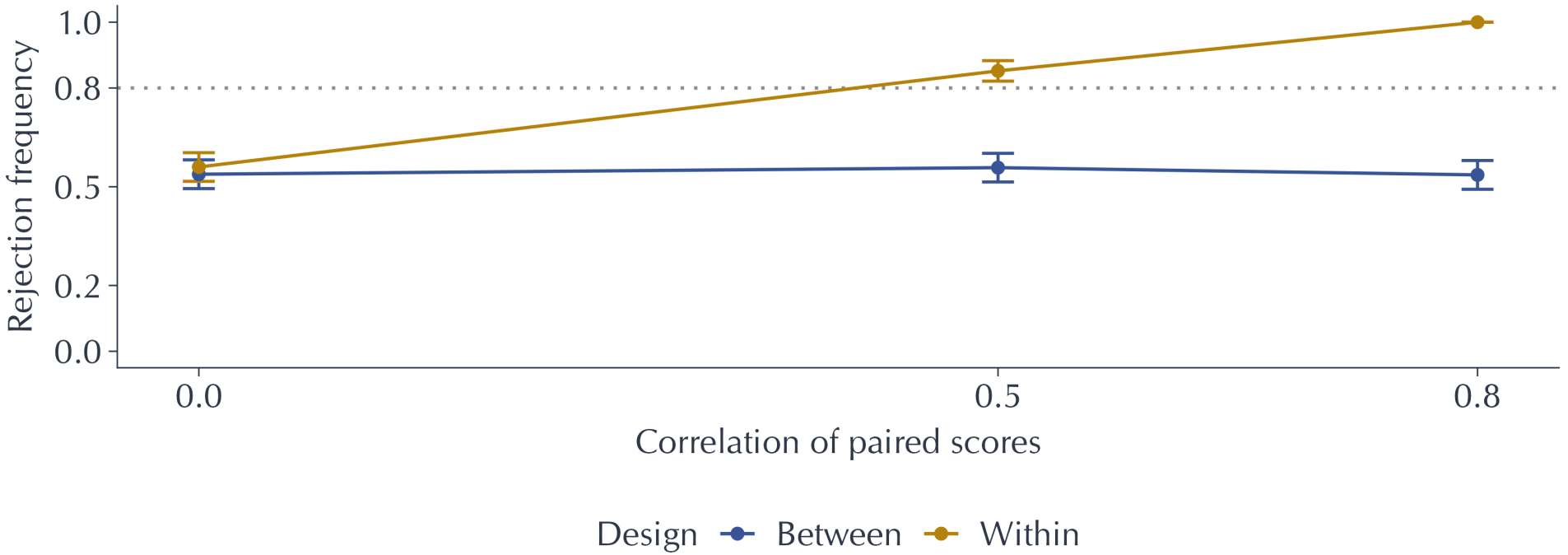
# Declare paired scores and estimate their differences

```r
model <- declare_model(
  N = 100, a = rnorm(N, sd = 10 * sqrt(rho)),
  Y_A = 50 + a + rnorm(N, sd = 10 * sqrt(1 - rho)),
  Y_B = 53 + a + rnorm(N, sd = 10 * sqrt(1 - rho))
)
inquiry <- declare_inquiry(ATE = 3)
measurement <- declare_measurement(D = Y_B - Y_A)
```

- **Estimate:** `mean(D)`.
  - Standard error: `sd(D) / sqrt(N)`.
  - Use a t interval and test.
- **Target:** population mean difference across units and runs.

# Pairing helps when the shared component is large

```
1 diagnose_design(make_paired(rho = .8), sims = 500)
```




- **Fixed:** 200 outcomes; $\tau = 3$; marginal SD = 10 points.
- **Vary `rho`:** how does comparability change power?

# Carryover breaks the two-Bundle SUTVA assumption

- **SUTVA:** no interference or hidden treatment versions.
  - A after B can differ from A with no prior exposure.
  - The current Bundle alone no longer defines the outcome.

$$Y_{it} = \mu + \alpha_i + \tau Z_{it} + \delta\, 1(t = 2) + \lambda Z_{i,t-1}.$$

- $\tau$: fresh-exposure effect; $\delta$: common period effect.
- $\lambda$: **carryover** from previous treatment.
  - People retain learning; model runs retain context or tool state.

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- $\lambda$: **carryover** from previous treatment.
  - People retain learning; model runs retain context or tool state.
- **Consequence:** the paired contrast can miss the fresh-exposure effect.

# Counterbalancing leaves carryover bias

- **A:** comparison Bundle. **B:** treatment Bundle.

| Randomized sequence | Expected B − A |
|---|---|
| A then B | $\tau + \delta$ |
| B then A | $\tau - \delta - \lambda$ |

- **Balance the sequences:** equally many people in each.

$$\mathrm{E}[\widehat{\tau}] = \tau - \lambda/2.$$

- Common period effect ($\delta$) cancels.
- Carryover leaves bias: $-\lambda/2$.
  - Randomizing order does **not** remove it.

# Within designs can be precise but biased

| Design | Bias | SD of estimates | 95% coverage |
|---|---|---|---|
| Crossover: carryover 0 | -0.077 | 1.020 | 0.938 |
| Crossover: carryover 3 points | -1.605 | 0.978 | 0.640 |
| Crossover: carryover 6 points | -2.999 | 0.991 | 0.156 |
| Parallel groups | 0.005 | 1.385 | 0.958 |

- **Coverage:** fraction of 95% intervals containing the target.
- **Target:** fresh-exposure effect, $\tau = 3$; 200 outcomes.
- **People:** 100 in crossover; 200 in parallel groups.
- **Tradeoff:** lower variance can come with bias and poor coverage.



# Blocking

# Blocking creates comparisons among similar people

- **Form blocks before assignment:** group people with similar pretest scores.
  - Sort 80 people by score; split into 4 groups of 20.
  - $b(i)$: person's block; $\gamma_{b(i)}$: shared baseline component.
  - The best thing to block on is the *prognostic score*, a prediction of $Y(0)$.

$$Y_i(0) = \mu + \gamma_{b(i)} + \epsilon_i, \qquad Y_i(1) = Y_i(0) + \tau.$$

```
assignment <- declare_assignment(
  Z = block_ra(blocks = block)
)
```

- **Randomize within each block:** 10 people per Bundle.
- **Estimate:** average the block contrasts, weighted by block size.

Gerber & Green (2012), §§3.6.1, 4.4.

# A special case gives a predictiveness threshold

- **Assume:** fixed population and constant effects.
  - J equal-sized blocks; same treatment fraction in every block.

$$\frac{V_{\text{blocked}}}{V_{\text{complete}}} = (1 - R_B^2)\frac{N-1}{N-J}.$$

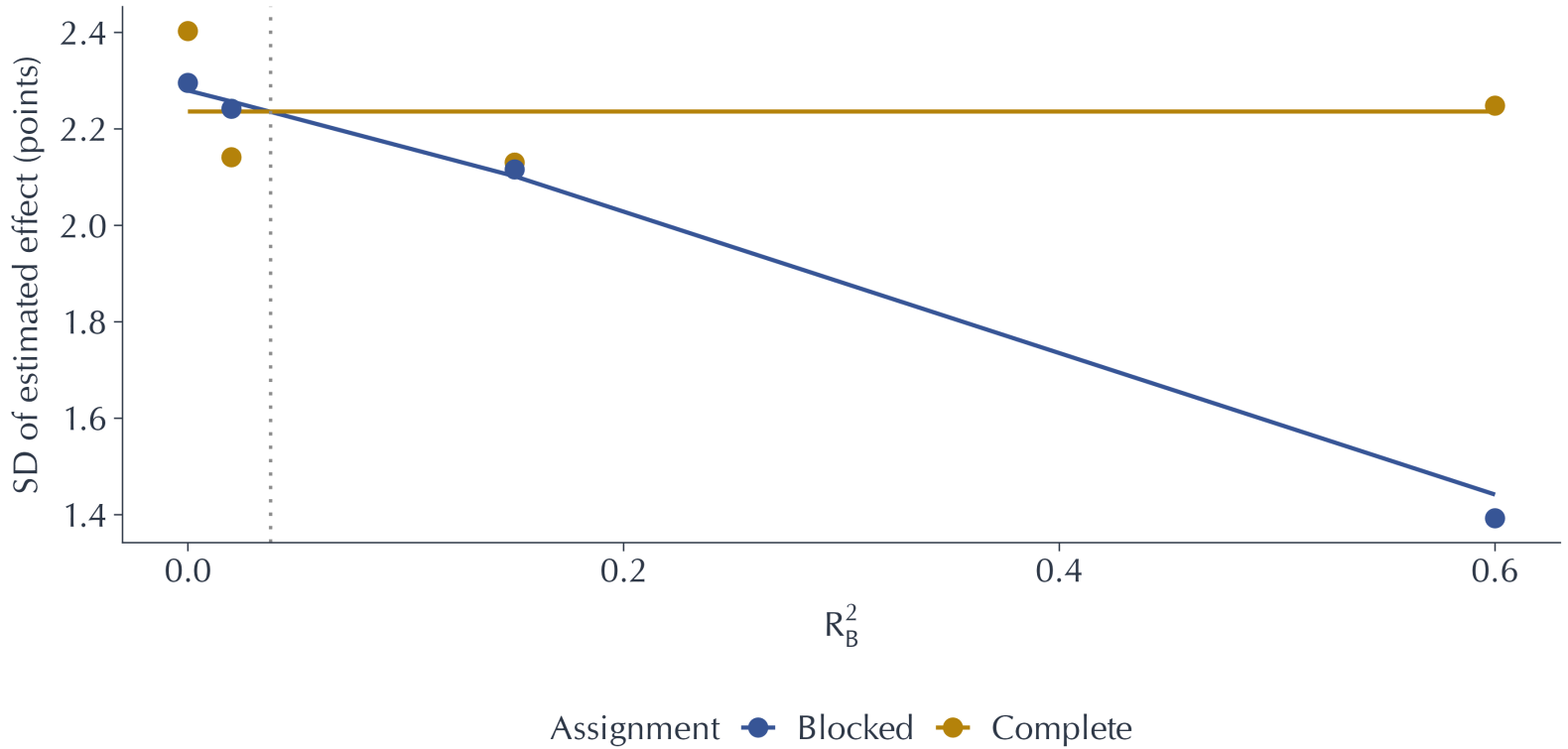
- **Blocking improves precision when:**

$$R_B^2 > \frac{J-1}{N-1}.$$

- $R_B^2$: share of $Y(0)$ variation explained by block indicators.
- $N = 80$, $J = 4$: threshold = **3.8%**.

Gerber & Green (2012), §3.6.1: variance comparison with complete randomization, specialized to equal blocks and constant effects.

# Prognostic blocks reduce assignment variability

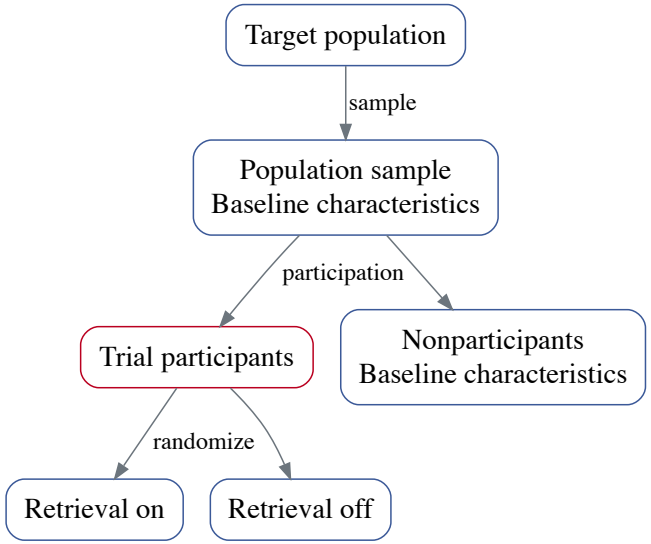



- **Lines:** analytic SD. **Points:** simulated SD.
- **Simulation:** fixed population; repeated assignment.
- **Choose blocks:** use pretreatment information or pilot evidence.

# Sampling and the evaluation budget

# Which population should the comparison inform?

- **So far:** how to compare Bundles. **Now:** which users and questions to study.
- **Human study target:** students who could use the tutor.




- **Sampling and participation**: who enters the study.
- **Random assignment**: which Bundle they receive.
  - Randomization alone does not make participants representative.

Adapted from Dahabreh et al. (2021), *Study designs for extending causal inferences from a randomized trial to a target population.*

# Questions and prompt phrasings are populations too

- **Mizrahi et al.:** changing instruction wording can change model rankings.
  - One wording gives a narrow view of performance.

| Sample dimension | Example target population |
|---|---|
| Questions | New questions covering the course's topics and difficulty range |
| Prompt phrasings | Ways intended users might ask those questions |

Mizrahi et al. (2024), *State of What Art?* Next week: Miller (2024), *Adding Error Bars to Evals.*



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  - If instruction is a treatment factor, sample phrasings within each level.
  - Paraphrases from a generator need not match real users' wording.

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  - If instruction is a treatment factor, sample phrasings within each level.
  - Paraphrases from a generator need not match real users' wording.
- **Claim:** an average over a stated population of possible questions and users.
  - More seeds on the same questions cannot fix missing task types.

Mizrahi et al. (2024), *State of What Art?* Next week: Miller (2024), *Adding Error Bars to Evals.*



# Paired comparisons still vary across questions

- **Model evaluation:** retrieval on vs. off, on the same sampled questions.
  - Fresh runs; score the model's answers. Hold instruction fixed.
- **Target:** average retrieval effect across possible course questions.

$$D_{ir} = Y_{ir}(B) - Y_{ir}(A) = \tau + u_i + e_{ir}.$$

- $u_i$: retrieval helps some questions more than others.
- $e_{ir}$: run-to-run noise in the paired score difference.

$$V(\bar{D}) = \frac{\sigma_u^2}{I} + \frac{\sigma_e^2}{IR}.$$

- $I$ **questions:** more questions reduce both terms.
- $R$ **runs per Bundle per question:** repeats reduce only run noise.
  - Average repeats within questions; estimate uncertainty across questions.

# Spend the same budget on questions or repeated runs

- **Fix:** two Bundles, a 3-point effect, and a budget of **1,200 cost units**.
- **Choose:** $R = 1, 2, 5$, or 10 runs per Bundle per question.
  - Use the remaining budget to maximize the number of questions $I$.

$$I = \left\lfloor \frac{1200}{c_{\text{question}} + 2R} \right\rfloor.$$

- **Run cost:** 1, including scoring. **Question preparation:** 0 or 10.
  - At cost 10: **100 questions × 1 run** or **40 questions × 10 runs**.

# Spend the same budget on questions or repeated runs

- **Fix:** two Bundles, a 3-point effect, and a budget of **1,200 cost units**.
- **Choose:** $R = 1, 2, 5$, or 10 runs per Bundle per question.
  - Use the remaining budget to maximize the number of questions $I$.

$$I = \left\lfloor \frac{1200}{c_{question} + 2R} \right\rfloor.$$

- **Run cost:** 1, including scoring. **Question preparation:** 0 or 10.
  - At cost 10: **100 questions × 1 run** or **40 questions × 10 runs**.
- **Two possible sources of variation:**
  - Mostly question differences: $(\sigma_u, \sigma_e) = (8, 6)$.
  - Mostly run noise: $(\sigma_u, \sigma_e) = (2, 10)$.

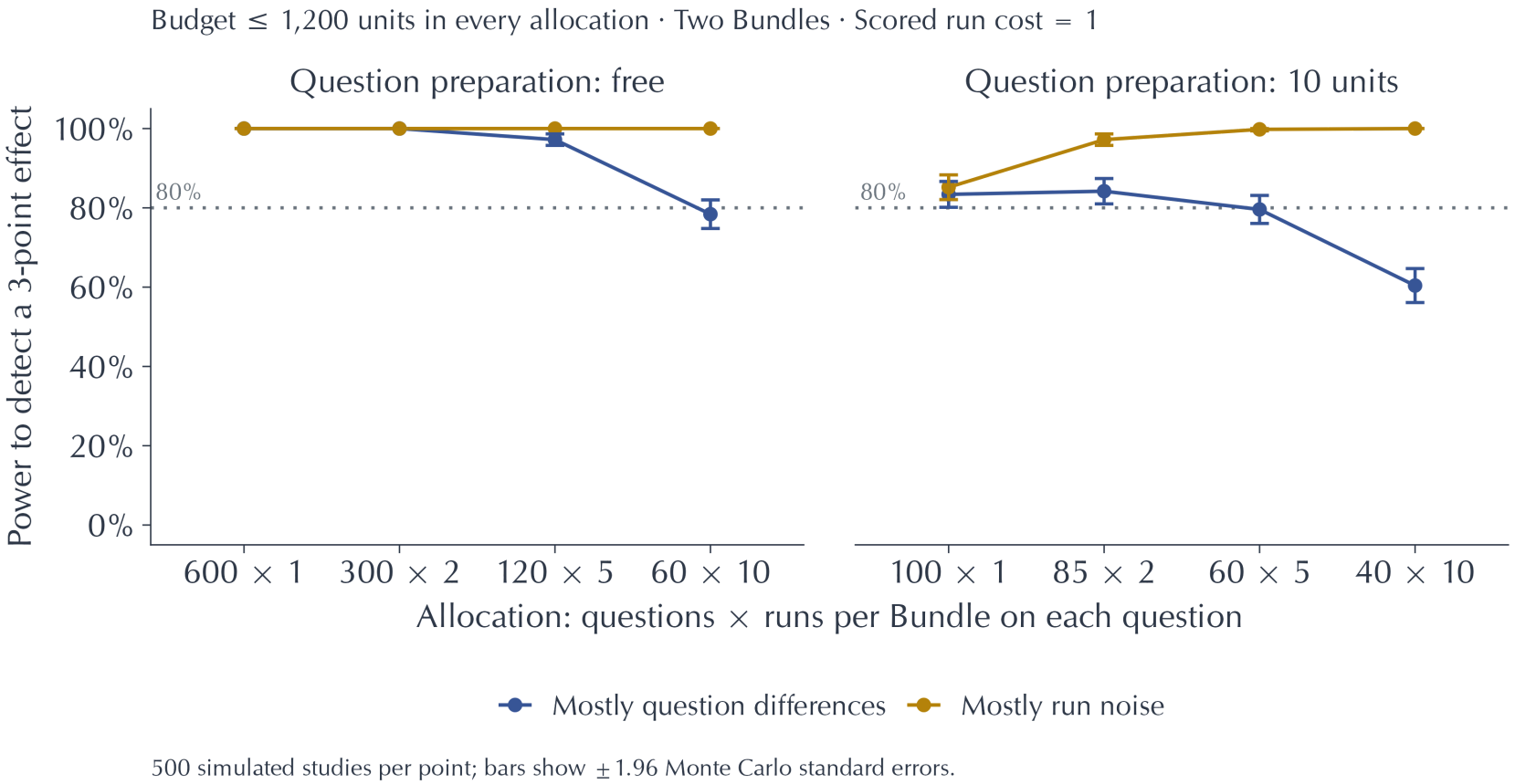
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- **For each allocation:** simulate 500 studies.
  - Power = fraction detecting the effect with a two-sided 5% test.

# How much power can the same budget buy?




- **Costly questions + noisy runs:** repeats can help; fewer questions may suffice.
- **Effects vary across questions:** too many repeats sacrifice power.

# Workshop preparation

# Bring your design decisions to next week's workshop

- **Build on your Bundle and comparison:** lock in the estimand between them.
- **Specify sampling:** who or what, from which population, and how.
- **Bring an initial power analysis.**
  - Use plausible measurement and design assumptions.
  - Start from this week's worked example.
  - Vary assumptions and design choices; see what you could detect.

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- **Together:** discuss these decisions and what else to consider.
- **Power analysis is approximate:** its answers depend on your assumptions.
  - Use it to think quantitatively about the consequences of design decisions.

Think of it like a **structure for clearer thinking.**

